Science of the Total Environment 902 (2023) 166053

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

The method matters. A comparative study of biologging and camera traps as data sources with which to describe wildlife habitat selection

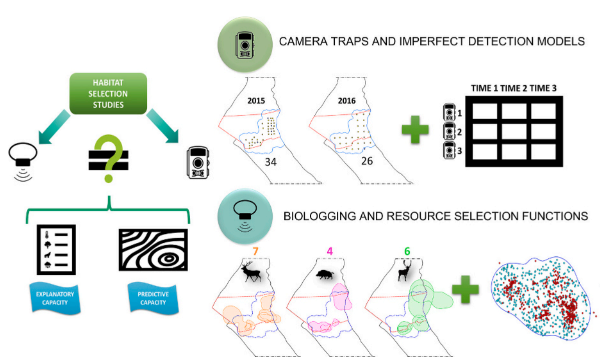
David Ferrer-Ferrando [a], Javier Fernández-López [b,c,*], Roxana Triguero-Ocaña [a], Pablo Palencia [a,d], Joaquín Vicente [a], Pelayo Acevedo [a,*]

[a] Instituto de Investigación en Recursos Cinegéticos (IREC), CSIC-UCLM-JCCM, Ciudad Real, Spain
[b] Université Montpellier, CNRS, EPHE, IRD, Montpellier, France
[c] Universidad Complutense de Madrid, Madrid, Spain
[d] Università Degli Studi di Torino, Dipartamiento di Scienze Veterinarie, Largo Paolo Braccini, 2, 10095 Grugliasco, Torino, Italy

HIGHLIGHTS

- We check if camera trap and biologging could describe habitat selection different.
- Wildlife habitat selection models consider land use and water related predictors.
- Each method identifies different features of the species' habitat use.
- Camera traps obtain information at population level and biologging at individual.
- We can combine both to obtain better models, or choose one based on our objective.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Martin Drews

Keywords:
N-mixture models
Resource selection functions
Imperfect detection models
Doñana National Park

ABSTRACT

Habitat use is a virtually universal activity among animals and is highly relevant as regards designing wildlife management and conservation actions. This has led to the development of a great variety of methods to study it, of which resource selection functions combined with biologging-derived data (RSF) is the most widely used for this purpose. However this approach has some constraints, such as its invasiveness and high costs. Analytical approaches taking into consideration imperfect detection coupled with camera trap data (IDM) have, therefore, emerged as a non-invasive cost-effective alternative. However, despite the fact that both approaches (RSF and IDM) have been used in habitat selection studies, they should also be comparatively assessed. The objective of

Abbreviations: RSF, resource selection functions combined with biologging-derived data; IDM, imperfect detection models coupled with camera trap data; DNP, Doñana National Park; BR, Biological Reserve; CR, Calibration Region; dwat, Euclidean distance (km) to the nearest artificial water hole; dvera, Euclidean distance (km) to the nearest marsh–shrub ecotone; v1, Proportion of dense scrub dominated by Erica scoparia and *Pistacia lentiscus*; v2, Proportion of low-clear shrubland, composed mainly of Halimium halimifolium, *Ulex minor* and Ulex australis; v3, Proportion of herbaceous grassland; v4, Proportion of Eucaliptus sp. and Pinus sp. Woodland; v5, Proportion of bare land, sandy dunes and beaches; v6, Proportion of watercourse vegetation covered mainly by Juncus sp. Patches; time, Sampling occasions of five days, the occasion date; year, Period date, in this case 2015 or 2016.; typeuse, Type of land use that predominates at the sampling point..

* Corresponding author at: SaBio Group-IREC, Ronda de Toledo, 12, 13005 Ciudad Real, Spain.
E-mail addresses: david.ferrer@uclm.es (D. Ferrer-Ferrando), javfer05@ucm.es (J. Fernández-López), joaquin.vicente@uclm.es (J. Vicente), pacevedo@irec.csic.es (P. Acevedo).

https://doi.org/10.1016/j.scitotenv.2023.166053
Received 14 April 2023; Received in revised form 10 July 2023; Accepted 2 August 2023
Available online 4 August 2023

Habitat use
Wild ungulates

this work is consequently to assess them from two perspectives: explanatory and predictive. This has been done by analyzing data obtained from camera traps (60 sampling sites) and biologging (17 animals monitored: 7 red deer *Cervus elaphus*, 6 fallow deer *Dama dama* and 4 wild boar *Sus scrofa*) in the same periods using IDM and RSF, respectively, in Doñana National Park (southern Spain) in order to explain and predict habitat use patterns for three studied species. Our results showed discrepancies between the two approaches, as they identified different predictors as being the most relevant to determine species intensity of use, and they predicted spatial patterns of habitat use with a contrasted level of concordance, depending on species and scale. Given these results and the characteristics of each approach, we suggested that although partly comparable interpretations can be obtained with both approaches, they are not equivalent but rather complementary. The combination of data from biologging and camera traps would, therefore, appear to be suitable for the development of an analytical framework with which to describe and characterise the habitat use processes of wildlife.

1. Introduction

Habitat use is an almost universal activity among animals and affects all of the individuals' choices and movement parameters (Begon et al., 2006; Manly et al., 2002). Its evident relevance has led considerable attention to be paid to both establishing a theoretical framework for habitat use studies and describing the ways in which organisms of different taxa actually evaluate and select from available habitats (Cody, 1981). One of the main conceptual bases of habitat use studies is that organisms should respond positively to environments in which their survival and reproductive success are higher (Levins, 1968; Orians, 1980). However, it is often difficult to establish correlations between habitat features and animals' fitness in real-world situations, since they are the result of multiple biotic and abiotic interactive factors that should be disentangled from individual/population monitoring data.

Biologging is (in a broad sense) the approach most frequently used to describe the habitat use patterns of wildlife (for a review, see Wilmers et al., 2015). Biologging consists of collecting remote data concerning free-ranging animals by using attached electronic devices (Cooke et al., 2004). This provides valuable information on the animals' movements and behaviour, which is very useful as regards understanding their spatial use and habitat selection patterns (Miller et al., 2010; Mulero-Pázmány et al., 2015). Biologging is, therefore, usually combined with resource selection functions in order to study habitat use patterns (Gillies et al., 2006). Briefly, the objective of these functions is to identify and parameterize the differences (in environmental terms) between animals' locations (*used)* and their *availability* in the area (Gillies et al., 2006; Manly et al., 2002). However, biologging still has some constraints, which are principally related to: i) its invasiveness, since it is necessary to capture animals, which requires a relevant sampling effort and could also potentially affects the animals' behaviour, and ii) its expensiveness, signifying that the budgets of most of research projects allow only a reduced number of animals to be monitored (Miller et al., 2010; Recio et al., 2011).

Camera traps are a non-invasive sampling method with a huge potential in wildlife monitoring (Burton et al., 2015; Iannarilli et al., 2021; O'Connell et al., 2011; Steenweg et al., 2017). They consist of automatically-trigger cameras that allow to collect photographic evidence of presence of animals in determined sites (Rovero et al., 2010; O'Connell et al., 2011). When non-invasive camera traps are coupled with statistical models that take into account imperfect detection, the result is a cost-effective alternative to biologging with which to study habitat use patterns (MacKenzie et al., 2002), assuming that the fitness of a species (dictated by habitat features) is correlated with population density (Boyce et al., 2016). These models, when employed in a hierarchical framework, use data obtained from sequential repeated (remote camera) surveys to generate probabilities of detection and to produce reliable estimates of the species' occupancy and abundance, in addition to determining the main drivers of these patterns (Kelly and Holub, 2008; MacKenzie and Royle, 2005; Royle, 2004). Nevertheless, camera traps have some constraints too: i) the time spent in processing photographic images (Jiménez et al., 2017; but see Delisle et al., 2021), the fact that data can only be obtained when animals are active, missing inactivity periods (Gould et al., 2019), and, when combined with N-mixture models, inability to know the effective sampling area of cameras (Gilbert et al., 2021).

Resource selection functions coupled with biologging data (hereafter denominated as the RSF approach), and imperfect detection models coupled with camera traps (hereafter denominated as the IDM approach) have been used together in various studies to, for example, describe population dynamics (Duquette et al., 2014), assess the transferability of inferences from the individual to the population level (Bassing et al., 2022) or design biological corridors (Meyer et al., 2020), among others. These approaches have been also used separately in order to describe wildlife habitat use (e.g. Goulart et al., 2009; Schofield et al., 2009; but see Coleman et al., 2014), although few comparative studies exploring the equivalence of the patterns described have been carried out (but see Bassing et al., 2022). However, it should be borne in mind that these approaches are based on two quite different sampling strategies, since biologging usually collects a lot of information from a few individuals (individual scale), while camera trapping collects (usually random) information from the different individuals present in specific points (population scale), so the first is limited with the number of individuals marked, and the second limited with the number of sites sampled. In addition, one must also be aware that the analytical processes are different, where resource selection functions analyse habitat use based on individual animals, while N-mixture models analyse the relative abundance of each site and associate it with habitat predictors. In both approaches, the obtained predictions can be interpreted as intensity of habitat use.

In this context, the objective of this study was to compare RSF and IDM when used in order to carry out wildlife studies on habitat use patterns. We have specifically worked with three highly-mobile mammal species: red deer (*Cervus elaphus*), fallow deer (*Dama dama*) and wild boar (*Sus scrofa*), and assessed whether both approaches are able to: i) identify the main environmental gradients explaining species intensity of habitat use and ii) produce spatial patterns of intensity of habitat use. Our hypothesis is that the inherent peculiarities of each approach (i.e. sampling and analytical differences) should lead to a description of habitat use on a different scale and that the results obtained after employing the two approaches will not, therefore, be equivalent.

2. Material and methods

2.1. Study area

The study was performed in Doñana National Park (hereafter, DNP; 37° 08′ N; 6° 47′ W), a nature reserve located on the Atlantic coast of South-Western Spain (Fig. 1). DNP has a total area of 54,252 ha, and hosts a variety of ecosystems including marshlands, lagoons, scrub woodland, forests and sand dunes, which has led to its declaration as a World Heritage Site and Biosphere Reserve (UNESCO 2014). This environmental heterogeneity maintains a great biodiversity. The group of wild ungulates in DNP includes a moderate density of red deer (6.3 individuals/100 ha, standard deviation [SD] 1.48) and fallow deer (3.9

individuals/100 ha, SD 0.99) and a moderately-high density of wild boar (5.7 individuals/100 ha, SD 1.18) (Vicente et al., 2014). DNP is characterised by the fact that it has rainy autumns and winters, and hot and dry summers, all of which produce irregular inlets of water that determine the ungulates' activity (Barasona et al., 2014a; Laguna et al., 2018). The Biological Reserve (BR) in the DNP, which is located in its central region (see the red polygon in Fig. 1), is the principal management area (related to cattle production) that overlaps with our data, while the study area considered herein was the Calibration Region (CR), in which sampled wild ungulate movement patterns overlapped, using biologging and camera traps (blue polygon in Figs. 1 and 2). CR was employed as a study area (because all habitat conditions are within environmental domain of the models, see Fig. 1S), while BR and DNP were used to extrapolate the prediction (because they have habitat conditions not necessarily within the domain of the model, as top right and left pictures show in Fig. 1S).

2.2. Environmental predictors

We selected the environmental predictors according to previous studies on wild ungulates' habitat use in our study area (Barasona et al., 2014a; Laguna et al., 2018; Triguero-Ocaña et al., 2020a): Euclidean distance (km) to the nearest artificial water hole (*dwat*); Euclidean distance (km) to the nearest marsh–shrub ecotone (*dvera*); proportion of dense scrub dominated by *Erica scoparia* and *Pistacia lentiscus* (*v1*); proportion of low-clear shrubland, composed mainly of *Halimium halimifolium*, *Ulex minor* and *Ulex australis* (*v2*); proportion of herbaceous grassland (*v3*); proportion of *Eucaliptus* sp. and *Pinus* sp. woodland (*v4*); proportion of bare land, sandy dunes and beaches (*v5*), and proportion of watercourse vegetation covered mainly by *Juncus* sp. patches (*v6*). For further descriptions of the layers, see Barasona et al. (2014a). The original data were raster layers of 10 × 10 meters for land use (*v1-v6*) and 100 × 100 meters for distances (*dvera* and *dwat*), but all were rescaled to 100 × 100 meter layers. These environmental predictors were estimated for each 100 × 100 meter grid that covers all the non-flooded region of DNP.

2.3. Resource selection functions and biologging data (RSF approach)

We used data concerning seven red deer, six fallow deer and four wild boar monitored with GPS-GSM collars (Microsensory System, Spain) from the second half of September to the first half of December 2015, and from the second half of March to the first half of April 2016. These animals were captured in the scrubland-marsh ecotone ("la vera"). The captures were carried out by a specialised scientist (B and C experimentation categories) following the protocol approved by the Animal Experiment Committee of Castilla-La Mancha University and by the Spanish Ethics Committee (PR-2015-03-08; for further details, see Triguero-Ocaña et al., 2020a). GPS-collars were set up to record one location every 2 h, and had a mean positioning error of 26 m (SD = 23.5 m). For each location, the collars also recorded the individuals' IDs, the date and the time (solar time). In order to obtain more similar and equivalent data to that of the IDM, we restricted the locations to the period of animal activity. Camera traps can detect animals only when they are active (Rowcliffe et al., 2014), while biologging provides data from all hours of the day. The periods of activity of the species in the study area (see Triguero-Ocaña et al., 2020b) were used as the basis on which to eliminate the central hours of the day (from 9:00 to 19:00 h) in order to ensure that both data sources were as comparable as possible.

Habitat use was assessed from biologging data by employing within-home-range resource-selection functions (Manly et al., 2002). The environmental information for each location considered in the RSF approach was assigned by using zonal statistics with the "extract" function from the "raster" R package (Hijmans, 2020), in this case considering a buffer of a 26 m radius around each one (according to GPS positional error; see also Triguero-Ocaña et al., 2020a). We compared the *used* versus *available* resources by using logistic regression mixed models (Gillies et al., 2006), in which the individual was included as a random-effect factor, and environmental characteristics were fixed factors. The use of resources was determined by the locations of each individual, while the availability was sampled by randomly creating 10 times the number of locations (1:10 ratio of used to available points; Fieberg et al., 2021) within each individual kernel home range 95 % (Khr95). We assigned a weight of 5000 to the available locations and a weight of 1 to the used locations (Fieberg et al., 2021). All these analyses were performed using R v.3.6.2 (R-Core Team, 2019), with the "adehabitat" (Calenge, 2006), and "glmmTMB" (Brooks et al., 2017) R packages.

2.4. Imperfect detection models and camera trap data (IDM approach)

Simultaneously to the biologging study (from the second half of September to the first half of December 2015, and from the second half of March to the first half of April 2016), 38 and 27 cameras (LTL Acorn, LTL-5310 series) were placed in the study area in the 2015 and 2016 seasons, respectively, of which 4 and 1 cameras were discarded from the analyses owing to operational problems (see Fig. 2). The cameras were set on wooden stakes between 30 and 50 cm above the ground, were

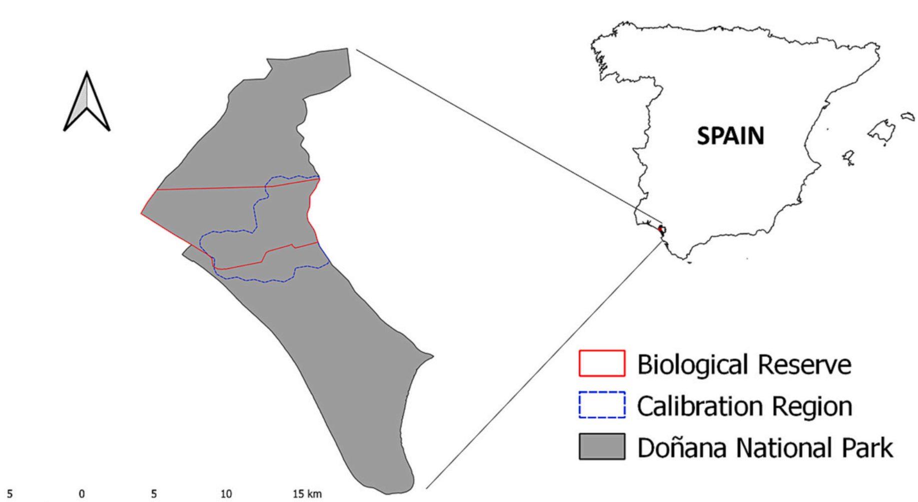


Fig. 1. Study area location. The Biological Reserve (principal management area that overlaps with our data; red continuous line) and the Calibration Region (our study area; blue discontinuous line) are located in Doñana National Park (grey area).

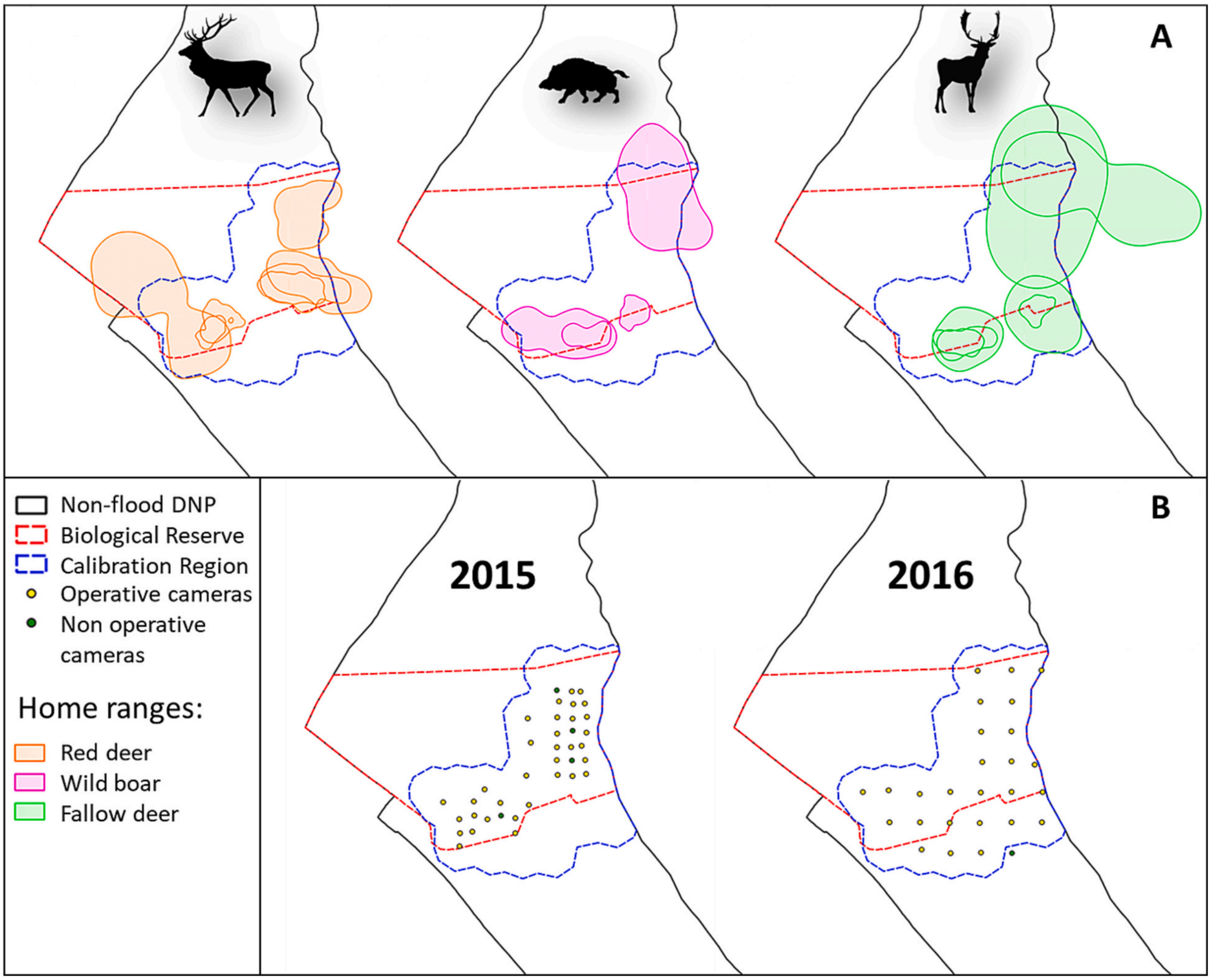


Fig. 2. Calibration Region (blue line) and Biological Reserve (red line), showing the home range (kernel 95 %) of collared individuals by species (A) and the grid of camera traps by sampling period (B).

programmed to record 3 consecutive pictures per activation, with <1 s between triggers and were active for 24 h. Camera traps have been established systematically with random origin, with a distance of 500 m between each of them in 2015 and 1 km in 2016, and no bait was used in either season.

We employed imperfect detection hierarchical models to determine habitat use with the data obtained from the camera traps (MacKenzie et al., 2002; Royle, 2004). The observational process (detectability) was estimated by splitting the study period into five-day occasions (sampling occasions) in order to avoid low detection probabilities (King et al., 2021), considering each occasion as a visit and each camera trap as a sampling unit (site). We additionally established a 10-minute window, such that detections >10 min apart were considered as independent events (Tanwar et al., 2021). Detectability may vary according to the site or survey characteristics, and we considered the occasion date (*time*) and the period date (*year*) as observation covariates, and the type of dominant land use (*typeuse*; i.e. type of land use that predominates at the sampling point) as a site covariate that could potentially affect the detection process. The ecological process (relative abundance) relates only to site characteristics, and we employed all the environmental predictors mentioned above as covariates of site. The environmental information concerning each of these variables was assigned to each camera trap using zonal statistics with the “extract” function from the “raster” R package (Hijmans, 2020). With regard to our study species (red deer, fallow deer and wild boar), we determined intensity of use by comparing the differences in the animals’ relative abundance around each camera trap (assuming that the fitness of a species is correlated with population density; Boyce et al., 2016), taking into account the environmental characteristics of each site by using counts of detections and single-season N-mixture models (Royle, 2004). As the focus was on spatial variation in habitat use, we used single-season models, while *year* was also included in the ecological process as a fixed factor in order to control for differences in intensity of use depending on the season. Another possible approach would have been the use of site-occupancy models (MacKenzie et al., 2002), which employ detection/non-detection histories (0/1) as input data. However, given the small size of our sampling area and the high average occupancy for two species (high percentage of study area occupied, see percentage of cameras with detections in Table 1S), we decided to use detection counts within an N-mixture modelling framework so as to incorporate as much as heterogeneity in habitat use for each species. Since it was not possible to meet the strict set of assumptions required by N-mixture models in order to obtain total abundance estimations (e.g. population sampled in closed units), we interpreted our model predictions as relative abundance or simply as intensity of habitat use (Searle et al., 2020). All analyses were performed using R v. 3.6.2 (R-Core Team, 2019), with the “unmarked” package (Fiske and Chandler, 2011).

2.5. Model selection and predictions

The best model for each approach was selected by following a backward stepwise selection procedure based on Akaike's information criteria (AIC). This procedure consists of starting with the full model (all predictor variables included), and in each step, removing the variable that most decreases the AIC score when removed. The stepwise procedure is stopped when it is not possible to improve the model by removing additional predictors. We maintained the model with lowest AIC and all the models with a similar fit (AIC difference lower than 2 units; Burnham and Anderson, 2002). In those cases in which more than one model was available, we create an average model based in these models, applying a weight to each model based on its AIC by using the "model.avg" function of "MuMIn" package (Burnham and Anderson, 2002).

Once the best model had been selected (one per species and approach), it was projected at a resolution of 100 × 100 meter in order to predict species intensity of habitat use in the CR by using the "predict" function from the "raster" R package (Hijmans, 2020). We comparatively assessed concordance between spatial patterns from each approach used for the CR, and also predicted the BR and DNP level so as to explore the consistence between approaches when the models were extrapolated (outside the domain of the model, see *dvera* and *dwat* in Fig. 1S). This was done by carrying out a reclassification in quartiles (0–25 %, 25–50 %, 50–75 % and 75–100 %) beforehand in order to avoid problems resulting from a different scale in the predictions, and led to the attainment of a common intensity of habitat use level category (levels from lowest to highest adequacy of the environment for the species). The agreement between the predictions was estimated by using a weighted Cohen's kappa coefficient (Cohen, 1960). The weighted kappa is a modification of Cohen's kappa that considers the closeness of agreement between categories when there are more than two, penalising the disagreement with greater force when the difference between categories is greater. The index ranges from −1 (complete disagreement) to 1 (complete agreement), and the value 0 indicates a concordance similar to that expected by chance. In our case, we employed a matrix of weights established as 0 on the diagonal and the distance from the diagonal squared outside of the diagonal (default conditions; Revelle, 2015).

3. Results

Both approaches, i.e. RSF and IDM, were compared in terms of the explanatory variables selected and the spatial patterns predicted. In the case of red deer, the main variables highlighted by the RSF approach were related to wet enclosed areas (by considering estimates *p*-value and z-value weights; see the model in Table 1). With regard to the IDM approach, the main variable was related to wet areas in the relative abundance process, and *year* and *time* in the detection process (Table 1). The agreement between the predictions generated by the two approaches (Fig. 3) was, according to Cohen's weighted kappa coefficient, 0.225 (with a confidence interval [CI] of 95 % from 0.181 to 0.270) at the CR level, 0.221 (CI95% from 0.210 to 0.235) at the BR level, and 0.112 (CI95% from 0.101 to 0.123) at the DNP level (Fig. 3).

In the case of wild boar, the main variable highlighted in the RSF approach was related to wet, dry and enclosed areas (Table 1). With regard to the IDM approach, the main variables were enclosed, dry and wet areas in the relative abundance process, and *year* in the detection process (Table 1). The agreement between the predictions generated by the two approaches for wild boar (Fig. 3) was, according to Cohen's weighted kappa coefficient, 0.260 (CI95% from 0.210 to 0.311) at the CR level, 0.490 (CI95% from 0.470 to 0.510) at the BR level, and 0.580 (CI95% from 0.570 to 0.590) at the DNP level (Fig. 3).

Finally, in the case of fallow deer, the main variables highlighted in the RSF approach were related to wet and enclosed areas (Table 1). With regard to the IDM approach, the main variables were related to enclosed and wet areas in the relative abundance process, and *year* in the detection process (Table 1). In this case, the agreement between predictions generated by the two approaches (Fig. 3) was, according to Cohen's weighted kappa coefficient, 0.150 (CI95% from 0.113 to 0.190) at the CR level, 0.360 (CI95% from 0.330 to 0.390) at the BR level, and 0.550 (CI95% from 0.540 to 0.550) at the DNP level (Fig. 3).

4. Discussion

Several comparative studies of different methodologies with which to study animal habitat use have been carried out (e.g. Mulero-Pázmány et al., 2015). Modern analytical approaches whose purpose is to take into account imperfect detection (such as site-occupancy or N-mixture models) have become popular in species occupancy and abundance modelling in recent years, and have consequently become complementary approaches to classical approaches already established in various research topics (e.g. Duquette et al., 2014; Meyer et al., 2020), one of which is the study of the habitat use (Coleman et al., 2014). The results obtained in this study suggest that RSF and IDM can produce partly comparable interpretations concerning intensity of habitat use, but that there are notable differences between them which vary according to species. The lack of support for equivalence between IDM and RSF approaches in our study support our hypothesis, but more studies using different populations, species, scales, contrasting environments and particularly longer periods of data collection should be carried out in order to verify our results. Moreover, these discrepancies could indicate that each approach captures different information on populations of the same species (Bassing et al., 2022), and a combination of both approaches could overcome the limitations of each one and improve the descriptions of wildlife habitat selection.

Our first objective was to discover whether the IDM approach was able to detect the main environmental gradients by determining intensity of habitat use in the same way that the RSF approach does. The tendency of both approaches to agree or not depends on the species. The common explanatory variables in the red deer models obtained for both approaches were those related to proximity to wet environments (*dvera* and *dwat*; see Table 1), but the RSF approach highlighted a preference for shrubland landscapes (*v2*, given that it shows a negative trend of all land uses, except *v2* that is missing), in addition to avoiding forest areas (*v4*; Table 1), which was what produced the difference between both approaches. All these tendences are consistent with previous studies carried out under Mediterranean climate conditions (e.g. Alves et al., 2014; Barasona et al., 2014a; Braza and Álvarez, 1987). The wild boar attained a negative relationship with shrubland (*v1* and *v2*) and drier areas (*v5*) for both approaches (Table 1), and this coincides with previous studies, which have shown that the wild boar avoids of shrubland during its activity period (Acevedo et al., 2006, 2014). Additionally, the RSF approach highlighted a clear tendency towards wet (*dwat* and *v6*) and herbaceous areas (*v3*; Table 1), and this dependence on water and herbaceous points in the driest season has also been shown in previous studies (Abaigar et al., 1994; Barasona et al., 2014b). Nevertheless, although RSF approach selected more environmental factors, both indicated the same trend, highlighting the water sources as a key factor in southern Spain. With regard to the fallow deer, both approaches coincide, showing a tendency to avoid more enclosed areas (*v1* and *v2*) and a strong preference for wet environments (*dwat* and *dvera*; Table 1), similar to the results obtained in previous studies (Braza and Álvarez, 1987). However, the RSF approach also included an avoidance of forest areas (*v4*, in IDM it was not statistically significant; Table 1), as has occurred in previous studies (Barasona et al., 2014a; Braza and Álvarez, 1987).

The RSF models were, in general, more complex for all the species (i.e. included a larger number of statistically significant predictors). This difference between approaches could be principally owing to the quantity and quality of the different sources of data: GPS collars record near continuous spatial data, no matter where the animal is moving, but at an individual level (Hebblewhite and Haydon, 2010), whereas camera

Table 1
Statistical parameters of the final habitat intensity of use models for red deer (*Cervus elaphus*), wild boar (*Sus scrofa*) and fallow deer (*Dama dama*) according to resource selection functions based on biologging-derived data (RSF approach) and imperfect detection models based on camera trapping data (IDM approach); *dvera* = Euclidean distance to nearest marsh-shrub ecotone, *dwat* = Euclidean distance to nearest artificial water hole, *v1* = proportion of dense scrub, *v2* = proportion of low-clear shrubland, *v3* = proportion of herbaceous grassland, *v4* = proportion of woodland, *v5* = proportion of bare land, sandy dunes and beaches, *v6* = proportion of watercourse vegetation, *time* = the occasion date, *typeuse* = type of land use that predominates at the sampling point, and *year* = a factor of two categories according to the year in which the data was recorded.

Species	Approach	Variable	Estimate	SE	Z	p-value
Red deer	RSF (Relative selection strength)	(Intercept)	6.409	53.884	0.107	0.915
		dwat	−0.140	0.023	6.11	< 2,00e-16
		dvera	0.013	0.032	0.41	0.681
		v1	−0.216	0.019	11.21	< 2,00e-16
		v3	−0.070	0.018	3.94	8.33e-05
		v4	−0.257	0.025	10.44	< 2,00e-16
		v5	−0.099	0.018	5.67	< 2,00e-16
		v6	−0.215	0.019	11.59	< 2,00e-16
		year	−0.015	0.038	0.39	0.697
Red deer	IDM (Relative Abundance)	(Intercept)	3.435	0.154	22.28	< 2,00e-16
		dwat	−0.215	0.133	1.63	0.104
		dvera	−0.474	0.115	4.11	3.90e-05
		v1	−0.027	0.118	0.23	0.819
		v3	−0.180	0.134	1.34	0.180
		v4	−0.045	0.112	0.40	0.691
		v5	−0.154	0.119	1.30	0.195
		v6	0.105	0.112	0.94	0.349
		year	0.371	0.426	0.87	0.384
	(Detection)	(Intercept)	−2.808	0.147	19.06	< 2,00e-16
		time	0.047	0.010	4.56	5.20e-06
		year	−1.397	0.397	3.52	4.31e-04
Wild boar	RSF (Relative selection strength)	(Intercept)	484.072	230.114	2.10	0.035
		dwat	−0.399	0.044	9.06	< 2,00e-16
		dvera	−0.059	0.090	0.65	0.513
		v1	−0.214	0.045	4.71	2.5e-06
		v3	0.235	0.039	6.10	< 2,00e-16
		v4	−0.079	0.041	1.92	0.055
		v5	−0.199	0.053	3.76	1.68e-04
		v6	0.339	0.037	9.20	< 2,00e-16
Wild boar	IDM (Relative abundance)	(Intercept)	2.390	0.325	7.35	< 2,00e-16
		dwat	−0.266	0.167	1.60	0.11038
		dvera	−0.288	0.167	1.72	0.08560
		v1	−0.545	0.387	1.41	0.15880
		v2	−1.135	0.465	2.44	0.01460
		v3	−0.680	0.432	1.57	0.11549
		v4	−0.098	0.229	0.43	0.66795
		v5	−0.738	0.300	2.46	0.01385
		v6	−0.442	0.328	1.35	0.17814
		year	−0.343	0.600	0.58	0.56560
	(Detection)	(Intercept)	−3.055	1.041	2.93	0.00335
		time	0.001	0.012	0.47	0.63816
		typeuse1	0.283	1.052	0.27	0.78777
		typeuse2	0.963	1.048	0.92	0.35819
		typeuse3	1.017	1.042	0.98	0.32872
		typeuse4	−0.618	1.380	0.45	0.65453
		typeuse5	1.870	2.174	0.86	0.38969
		typeuse6	−1.556	1.205	1.29	0.19662
		year	−1.438	0.560	2.56	0.01022
Fallow deer	RSF (Relative selection strength)	(Intercept)	−10.884	0.073	−149.62	< 2,00e-16
		dwat	−0.444	0.021	−21.29	< 2,00e-16
		dvera	−0.243	0.034	−7.17	7.40e-13
		v1	−0.541	0.033	−16.32	< 2,00e-16
		v2	−0.316	0.018	−17.29	< 2,00e-16
		v3	−0.220	0.017	−12.67	< 2,00e-16
		v4	−0.245	0.020	−12.24	< 2,00e-16
Fallow deer	IDM (Relative abundance)	(Intercept)	2.390	0.325	7.35	< 2,00e-16
		dwat	−0.221	0.143	1.54	0.124
		dvera	−0.808	0.141	5.73	< 2,00e-16
		v2	−0.524	0.168	3.11	1.89e-03
		v3	−0.669	0.158	4.23	2.32e-05
		v4	−0.228	0.178	1.28	0.200

(*continued on next page*)

Table 1 (*continued*)

Species	Approach	Variable	Estimate	SE	Z	p-value
		v5	−0.053	0.422	0.13	0.900
		year	1.718	1.427	1.20	0.228
	(Detection)	(Intercept)	−2.504	0.455	5.503	< 2,00e-16
		time	0.024	0.026	0.93	0.353
		year	−2.476	1.486	1.67	0.096

traps can obtain near continuous temporal detections of many individuals at a population level, but are limited to specific survey points (Burton et al., 2015; Iannarilli et al., 2021; O'Connell et al., 2011). Camera traps monitor finite space and locations, thus limiting the extent and resolution of inference and, therefore, the power to estimate the effects of multiple predictors at once (Bailey et al., 2007; Phillips et al., 2019; Wakefield et al., 2011; Watanuki et al., 2016). When using camera traps it is consequently necessary to employ a large number of sampling points in order to obtain the quantity of variation required to obtain intensity of habitat use precise predictions from complex models and avoid bias in the sampling process (Burton et al., 2015; Hofmeester et al., 2019; Iannarilli et al., 2021; Tanwar et al., 2021).

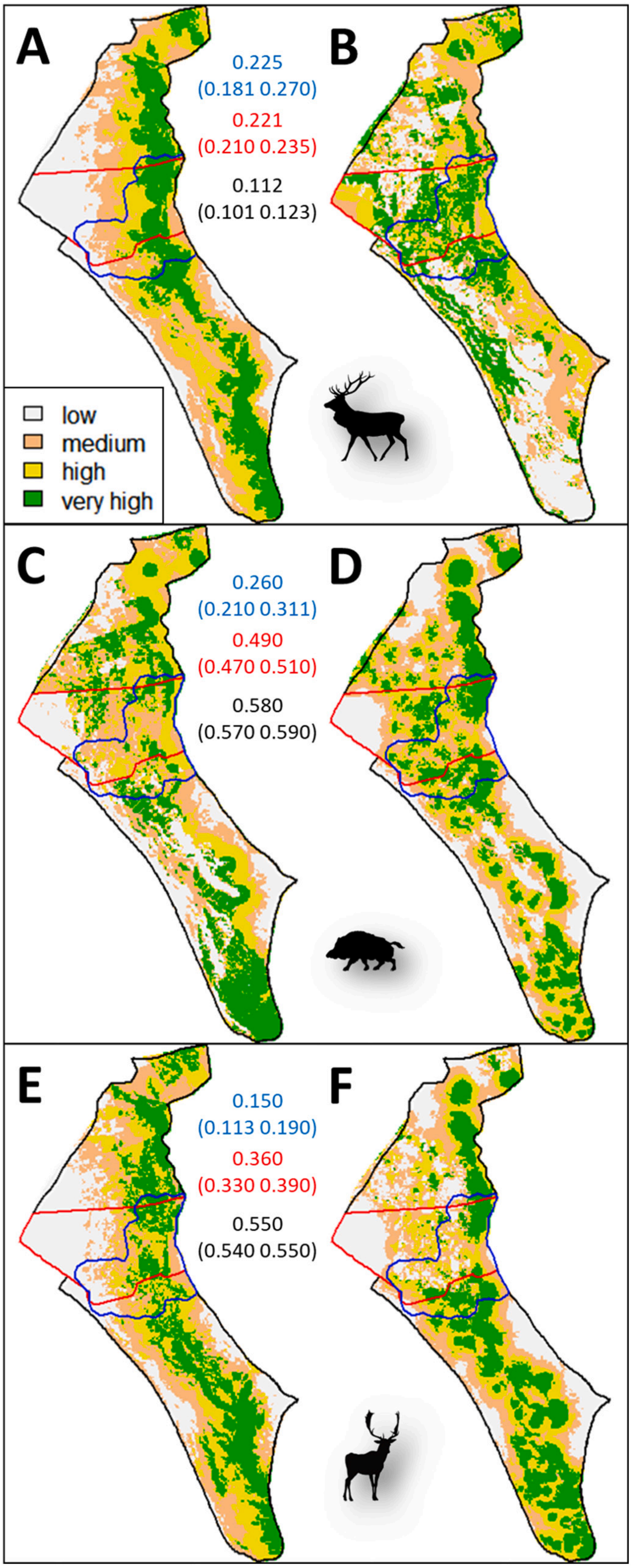
Our second objective was to explore the consistency of the predicted patterns of intensity of habitat use in each approach. In this respect, we obtained differences in consistence among the species at the three levels: CR, BR and DNP. These types of discrepancies in predictions have already been seen in previous studies (Bassing et al., 2022; Phillips et al., 2019). Concretely, agreement was obtained for all the species in the study area (CR level; see Fig. 3). The best concordance was observed for wild boar, followed by fallow deer and red deer (Fig. 3), which may be related to the similarity of the explanatory factors in the RSF and IDM approaches for each species (Table 1). Nevertheless, prediction agreement followed different trends when extrapolated (BR and DNP level), depending on the species, and agreement decreased when comparisons were made on larger scales (CR > BR > DNP; see the visual pattern and Kappa coefficient in Fig. 3) for red deer, while agreement increased with scale (DNR > BR > CR; Fig. 3) for wild boar and fallow deer. One possible explanation for this discrepancy could be related to the models' capacity to capture the complete environmental gradient of the species in the sampled area. When the more relevant gradients for the species are not included in the model, imprecise predictions could be produced in the study area, and higher discrepancies in new (extrapolated) territories (Elith and Leathwick, 2009; see differences in covariate gradients between methods and/or levels of prediction in Fig. 1S, as differences in quantity of data of typeuse *v1* and *v2* between approaches). According to this hypothesis, the models for red deer produce under-representative predictions (in one or both approaches), and the disagreement between them increases when they are extrapolated. In the case of wild boar and fallow deer, both predictions were able to capture the environmental characteristics of the preferred areas and, therefore, provided good descriptions of the general patterns on large spatial scales (increasing the agreement to these scales).

When the intensity of habitat use patterns obtained were compared with those shown in previously published studies, IDM approach for red deer was more concordant (Barasona et al., 2014a). In the case of wild boar and fallow deer, both approaches were similar to those shown in previous works (Barasona et al., 2014a, 2014b). One reason for the discordance between the red deer RSF model and those shown in previous works may be the short data-collection period employed and/or a scarce number of individuals monitored, which could have produced bias (caused by a lack of time and/or individuals sampling data). As a general recommendation, employing longer periods in both approaches, along with establishing more sampling points in the case of IDM and more individual in the case of RSF, could lead to more robust predictions, since it would be possible to provide a good characterisation of all the habitats and preferences (Bassing et al., 2022; Phillips et al., 2019).

Our study showed that each approach differed as regards the main variables that influenced intensity of habitat use, but that all the selected variables are, to some extent, refuted by previous research. This may indicate that both methodologies can correctly determine habitat use, but that each of them identifies different features (Bassing et al., 2022). This is something that could be expected if we look at the key peculiarities of each approach, where RSF collects data at the individual level and analyzes habitat use against availability within each animal's home range, while camera trapping works at the population level in specific sites, comparing the relative abundance between the different sampled points. Therefore, the RSF approach is relating the predictor variables to the relative probability of selection within home range, while the IDM approach is relating the predictor variables to relative abundance differences between sampled sites. One reason of discrepancies based on these key peculiarities is the multi-scale interaction of the species with the environment (McGarigal et al., 2016), signifying that each approach detects some scales better than others. This may be owing to these key peculiarities of each approach (concretely individual-specific data in RSF vs. site-specific data in IDM). The RSF approach may, therefore, focus more on the within home range habitat selection (3rd order of Johnson's four levels of habitat selection), while the IDM approach may focus on the home range habitat selection (2nd order of Johnson's levels), bearing in mind that the variables that affect each level are not necessarily the same (Bassing et al., 2022; Boyce, 2006; Cushman and McGarigal, 2004). Another reason could be that each approach obtains a different range of values for the predictors (see differences between approaches in ranges as *dvera* or *v6*; Fig. S1), sampling different components of the same population's habitat use (Bassing et al., 2022). In other words, one approach obtains more variability from some predictor variables, and the other obtains more variability from others, and the set of predictors with greater variability may have more weight in each model. This may be related to the limitations of each model, where the IDM only obtains data on the predictors in a finite number of sampling points, the RSF obtains them from a finite number of home range of monitored animals. According to the results obtained in the present work, the use of biologging as a classic tool in habitat use studies, and the increase in the use of camera trapping in the last decades as a novel tool (Rovero and Zimmermann, 2016), show that both methods are complementary, and that approach greatly depends on the objective of the study. The complementary information contained in each kind of data therefore indicates that data integration is a promising tool with which to obtain the more informative and precise models of habitat selection required in order to support decision-making in wildlife management and conservation (Apollonio et al., 2017; Fletcher Jr et al., 2019; Isaac et al., 2020; Miller et al., 2019).

5. Conclusions

The two approaches tested, i.e. RSF and IDM, were not equivalent as regards identifying the main environmental gradients that explain the intensity of habitat use of the species studied: red deer, wild boar and fallow deer. The spatial patterns of habitat use have a greater or a lower level of agreement depending on the species studied. The spatial patterns maintained (and sometimes increased) the agreement between approaches beyond the study area (Biological Reserve and Doñana National Park) as long as the predictions were representative of the complete environmental gradient. Both approaches determined intensity of



(caption on next column)

Fig. 3. Predicted patterns of habitat intensity of use for wild ungulates in the Calibration Region (blue line). The Biological Reserve (red line) and Doñana National Park have been delimited in order to extrapolate predictions. The approaches were resource selection functions obtained from biologging derived data (RSF approach) and imperfect detection models obtained from camera trap data (IDM approach). Maps represent IDM (A) and RSF approach (B) for red deer; IDM (C) and RSF approach (D) for wild boar; and IDM (E) and RSF approach (F) for fallow deer. Habitat intensity of use patterns were divided into four levels (quantiles) in order to assess the agreement between the predictions produced by the approaches with Cohen's weighted kappa coefficient at three different levels of extension, with their respective 95 % confidence interval values in brackets: Calibration Region (study area, in blue), Biological Reserve (in red) and Doñana National Park (in black).

habitat use patterns well, but identified different features of it, probably because IDM relates environmental features to differences in relative abundance to a population level, while RSF relates environmental features to the relative probability of use at individual level. If the need arises to choose between both methodologies, it is, therefore, necessary to consider which best fits the proposed objective, or whether it is possible to combine both, thus resulting in the attainment of more informative and more precise habitat selection models. Future studies should consider repeating this comparison of approaches with longer periods of data collection and on more patchy landscapes in order to obtain a more robust conclusion.

Funding

DFF received a Start to Research grant for Master's degree students, financed by the collaboration agreement between the University of Castilla-La Mancha and the Banco Santander (2019-UNIVERS-9556), which allowed him to commence this study, and he also obtained support from the UCLM through a predoctoral scholarship (2020-PREDUCLM-16022), which allowed him to continue with this research. JFL was funded by the Margarita Salas grant from the European Union – NextGenerationEU through the Complutense University of Madrid. PP received support from University of Castilla-La Mancha through a Margarita Salas contract (2022-NACIONAL-110053). Our research group is partly supported by Applied Research Projects (2022-GRIN-34227) funded by UCLM-FEDER.

CRediT authorship contribution statement

David Ferrer-Ferrando: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Javier Fernández-López:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – review & editing, Visualization, Supervision. **Roxana Triguero-Ocaña:** Investigation, Resources, Data curation, Writing – review & editing. **Pablo Palencia:** Investigation, Resources, Data curation, Writing – review & editing. **Joaquín Vicente:** Conceptualization, Investigation, Resources, Data curation, Writing – review & editing, Funding acquisition. **Pelayo Acevedo:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data employed for N-mixture models and resource selection functions are available in Zenodo at https://zenodo.org/record/8214538. Code employed in the analysis is available in Zenodo at https://zenodo.org/record/8214538. The following data supporting this research are sensitive and not available publicly: coordinates of telemetry relocations from GPS-collared animals and camera trap locations.

Acknowledgements

We would like to thank all those who have participated in the capture and collaring of wild ungulates, and in the setting of camera traps.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2023.166053.

References

Abaigar, T., Del Barrio, G., Vericad, J.R., 1994. Habitat preference of wild boar (Sus scrofa L., 1758) in a Mediterranean environment. Indirect evaluation by signs. Mammalia 58 (2), 201–210.
Acevedo, P., Escudero, M.A., Muñoz, R., Gortázar, C., 2006. Factors affecting wild boar abundance across an environmental gradient in Spain. Acta Theriol. 51 (3), 327–336.
Acevedo, P., Quirós-Fernández, F., Casal, J., Vicente, J., 2014. Spatial distribution of wild boar population abundance: basic information for spatial epidemiology and wildlife management. EcologicalIndicators 36, 594–600.
Alves, J., da Silva, A.A., Soares, A.M., Fonseca, C., 2014. Spatial and temporal habitat use and selection by red deer: the use of direct and indirect methods. Mamm. Biol. 79 (5), 338–348.
Apollonio, M., Belkin, V.V., Borkowski, J., Borodin, O.I., Borowik, T., Cagnacci, F., Gaillard, J.M., 2017. Challenges and science-based implications for modern management and conservation of European ungulate populations. Mammal Res. 62 (3), 209–217.
Bailey, L.L., Hines, J.E., Nichols, J.D., MacKenzie, D.I., 2007. Sampling design trade-offs in occupancy studies with imperfect detection: examples and software. Ecol. Appl. 17 (1), 281–290.
Barasona, J.A., Mulero-Pázmány, M., Acevedo, P., Negro, J.J., Torres, M.J., Gortázar, C., Vicente, J., 2014a. Unmanned aircraft systems for studying spatial abundance of ungulates: relevance to spatial epidemiology. PLoS One 9 (12), 1–17.
Barasona, J.A., Latham, M.C., Acevedo, P., Armenteros, J.A., Latham, A.D.M., Gortazar, C., Vicente, J., 2014b. Spatiotemporal interactions between wild boar and cattle: implications for cross-species disease transmission. Vet. Res. 45 (1), 122.
Bassing, S.B., Devivo, M., Ganz, T.R., Kertson, B.N., Prugh, L.R., Roussin, T., Gardner, B., 2022. Are we telling the same story? Comparing inferences made from camera trap and telemetry data for wildlife monitoring. Ecol. Appl. e2745.
Begon, M., Townsend, C.R., Harper, J.L., 2006. *Ecology: From Individuals to Ecosystems* (No. Sirsi) i9781405111171.
Boyce, M.S., 2006. Scale for resource selection functions. Divers. Distrib. 12 (3), 269–276.
Boyce, M.S., Johnson, C.J., Merrill, E.H., Nielsen, S.E., Solberg, E.J., Van Moorter, B., 2016. Can habitat selection predict abundance? J. Anim. Ecol. 85 (1), 11–20.
Braza, F., Álvarez, F., 1987. Habitat Use by Red Deer and Fallow Deer in Doñana National Park. Miscel·lània Zoològica, pp. 363–367.
Brooks, M.E., Kristensen, K., van Benthem, K.J., Magnusson, A., Berg, C.W., Nielsen, A., Bolker, B.M., 2017. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. R J. 9 (2), 378–400.
Burnham, K.P., Anderson, D.R., 2002. Model selection and multimodel inference. In: A Practical Information-theoretic Approach, 2nd ed. Springer, New York, p. 2.
Burton, A.C., Neilson, E.W., Moreira, D., Ladle, A., Steenweg, R., Fisher, J.T., Bayne, E. M., Boutin, S., 2015. REVIEW: wildlife camera trapping: a review and recommendations for linking surveys to ecological processes. J. Appl. Ecol. 52, 675–685.
Calenge, C., 2006. The package "adehabitat" for the R software: a tool for the analysis of space and habitat use by animals. Ecol. Model. 197 (3–4), 516–519.
Cody, M.L., 1981. Habitat selection in birds: the roles of vegetation structure, competitors, and productivity. BioScience 31 (2), 107–113.
Cohen, J.A., 1960. Educational and psychological measurement. In: Coefficient of Agreement for Nominal Scales, 20, pp. 37–46.
Coleman, L.S., Ford, W.M., Dobony, C.A., Britzke, E.R., 2014. Comparison of radio-telemetric home-range analysis and acoustic detection for little brown bat habitat evaluation. Northeast. Nat. 21 (3), 431–445.
Cooke, S.J., Hinch, S.G., Wikelski, M., Andrews, R.D., Kuchel, L.J., Wolcott, T.G., Butler, P.J., 2004. Biotelemetry: a mechanistic approach to ecology. Trends Ecol. Evol. 19 (6), 334–343.
Cushman, S.A., McGarigal, K., 2004. Patterns in the species–environment relationship depend on both scale and choice of response variables. Oikos 105 (1), 117–124.
Delisle, Z.J., Flaherty, E.A., Nobbe, M.R., Wzientek, C.M., Swihart, R.K., 2021. Next-generation camera trapping: systematic review of historic trends suggests keys to expanded research applications in ecology and conservation. Front. Ecol. Evol. 9, 617996.
Duquette, J.F., Belant, J.L., Svoboda, N.J., Beyer, D.E., Albright, C.A., 2014. Comparison of occupancy modeling and radiotelemetry to estimate ungulate population dynamics. Popul. Ecol. 56 (3), 481–492.
Elith, J., Leathwick, J.R., 2009. Species distribution models: ecological explanation and prediction across space and time. Annu. Rev. Ecol. Evol. Syst. 40 (1), 677–697.
Fieberg, J., Signer, J., Smith, B., Avgar, T., 2021. A 'how to' guide for interpreting parameters in habitat-selection analyses. J. Anim. Ecol. 90 (5), 1027–1043.
Fiske, I., Chandler, R., 2011. Unmarked: an R package for fitting hierarchical models of wildlife occurrence and abundance. J. Stat. Softw. 43 (10), 1–23.
Fletcher Jr., R.J., Hefley, T.J., Robertson, E.P., Zuckerberg, B., McCleery, R.A., Dorazio, R.M., 2019. A practical guide for combining data to model species distributions. Ecology 100 (6), e02710.
Gilbert, N.A., Clare, J.D., Stenglein, J.L., Zuckerberg, B., 2021. Abundance estimation of unmarked animals based on camera-trap data. Conserv. Biol. 35 (1), 88–100.
Gillies, C.S., Hebblewhite, M., Nielsen, S.E., Krawchuk, M.A., Aldridge, C.L., Frair, J.L., Jerde, C.L., 2006. Application of random effects to the study of resource selection by animals. J. Anim. Ecol. 75 (4), 887–898.
Goulart, F.V.B., Cáceres, N.C., Graipel, M.E., Tortato, M.A., Ghizoni Jr., I.R., Oliveira-Santos, L.G.R., 2009. Habitat selection by large mammals in a southern Brazilian Atlantic Forest. Mamm. Biol. 74 (3), 182–190.
Gould, M.J., Gould, W.R., Cain, J.W., Roemer, G.W., 2019. Validating the performance of occupancy models for estimating habitat use and predicting the distribution of highly-mobile species: a case study using the American black bear. Biol. Conserv. 234 (March), 28–36.
Hebblewhite, M., Haydon, D.T., 2010. Distinguishing technology from biology: a critical review of the use of GPS telemetry data in ecology. Phil. Trans. R. Soc. B Biol. Sci. 365 (1550), 2303–2312.
Hijmans, R.J., 2020. raster: geographic data analysis and modeling. R package version 3.3-6. https://CRAN.R-project.org/package=raster.
Hofmeester, T.R., Cromsigt, J.P., Odden, J., Andrén, H., Kindberg, J., Linnell, J.D., 2019. Framing pictures: a conceptual framework to identify and correct for biases in detection probability of camera traps enabling multi-species comparison. Ecol. Evol. 9 (4), 2320–2336.
Iannarilli, F., Erb, J., Arnold, T.W., Fieberg, J.R., 2021. Evaluating species-specific responses to camera-trap survey designs. Wildl. Biol. 2021 (1), 1–12.
Isaac, N.J., Jarzyna, M.A., Keil, P., Dambly, L.I., Boersch-Supan, P.H., Browning, E., O'Hara, R.B., 2020. Data integration for large-scale models of species distributions. Trends Ecol. Evol. 35 (1), 56–67.
Jiménez, J., Higuero, R., Charre-Medellin, J.F., Acevedo, P., 2017. Spatial mark-resight models to estimate feral pig population density. Hystrix. Ital. J. Mammal. 28 (2), 208–213.
Kelly, M.J., Holub, E.L., 2008. Camera trapping of carnivores: trap success among camera types and across species, and habitat selection by species, on salt pond mountain, Giles County, Virginia. Northeast. Nat. 15 (2), 249–262.
King, T.W., Vynne, C., Miller, D., Fisher, S., Fitkin, S., Rohrer, J., Thornton, D.H., 2021. The influence of spatial and temporal scale on the relative importance of biotic vs. abiotic factors for species distributions. Divers. Distrib. 27 (2), 327–343.
Laguna, E., Barasona, J.A., Triguero-Ocaña, R., Mulero-Pázmány, M., Negro, J.J., Vicente, J., Acevedo, P., 2018. The relevance of host overcrowding in wildlife epidemiology: a new spatially explicit aggregation index. Ecol. Indic. 84, 695–700.
Levins, R., 1968. Evolution in Changing Environments: Some Theoretical Explorations (No. 2). Princeton University Press.
MacKenzie, D.I., Royle, J.A., 2005. Designing occupancy studies: general advice and allocating survey effort. J. Appl. Ecol. 42 (6), 1105–1114.
MacKenzie, D.I., Nichols, J.D., Lachman, G.B., Droege, S., Royle, J.A., Langtimm, C.A., 2002. Estimating site occupancy rates when detection probabilities are less than one. Ecology 83 (8), 2248–2255.
Manly, B.F.J., McDonald, L.L., Thomas, D.L., McDonald, T.L., Erickson, W.P., 2002. Resource Selection by Animals: StatisticalAnalysis and Design for Field Studies, 2nd ed. Kluwer Academic Press, Boston, MA.
McGarigal, K., Wan, H.Y., Zeller, K.A., Timm, B.C., Cushman, S.A., 2016. Multi-scale habitat selection modeling: a review and outlook. Landsc. Ecol. 31 (6), 1161–1175.
Meyer, N.F.V., Moreno, R., Reyna-Hurtado, R., Signer, J., Balkenhol, N., 2020. Towards the restoration of the Mesoamerican biological corridor for large mammals in Panama: comparing multi-species occupancy to movement models. Mov. Ecol. 8 (1), 1–14.
Miller, C.S., Hebblewhite, M., Goodrich, J.M., Miquelle, D.G., 2010. Review of research methodologies for tigers: telemetry. Integr. Zool. 378–389.
Miller, D.A., Pacifici, K., Sanderlin, J.S., Reich, B.J., 2019. The recent past and promising future for data integration methods to estimate species' distributions. Methods Ecol. Evol. 10 (1), 22–37.
Mulero-Pázmány, M., Barasona, J.Á., Acevedo, P., Vicente, J., Negro, J.J., 2015. Unmanned aircraft systems complement biologging in spatial ecology studies. Ecol. Evol. 5 (21), 4808–4818.
O'Connell, A.F., Nichols, J.D., Karanth, K.U., 2011. In: O'Connell, A.F., Nichols, J.D., Karanth, K.U. (Eds.), Camera Traps in Animal Ecology: Methods and Analyses. Springer, New York, New York.
Orians, G.H., 1980. Habitat Selection: General Theory and Applications to Human Behavior. The Evolution of Human Social Behavior.
Phillips, E.M., Horne, J.K., Zamon, J.E., Felis, J.J., Adams, J., 2019. Does perspective matter? A case study comparing Eulerian and Lagrangian estimates of common murre (Uria aalge) distributions. Ecol. Evol. 9 (8), 4805–4819.
R Core Team, 2019. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Recio, M.R., Mathieu, R., Maloney, R., Seddon, P.J., 2011. Cost comparison between GPS-and VHF-based telemetry: case study of feral cats *Felis catus* in New Zealand. N. Z. J. Ecol. 114–117.
Revelle, W., 2015. Find Cohen's kappa and weighted kappa coefficients for correlation of two raters. R: Find Cohen's kappa and weighted kappa coefficients for. personality-project.org.
Rovero, F., Zimmermann, F., 2016. Camera Trapping for Wildlife Research. *Pelagic Publishing*, UK, Exeter.
Rovero, F., Tobler, M., Sanderson, J., 2010. Camera trapping for inventorying terrestrial vertebrates. In: Eymann, J., Degreef, J., Häuser, C., Monje, J.C., Samyn, Y., Vanden Spiegel, D. (Eds.), Manual on Field Recording Techniques and Protocols for All Taxa Biodiversity Inventories and Monitoring. The Belgian National Focal Point to the Global Taxonomy Initiative, 8, pp. 100–128.
Rowcliffe, J.M., Kays, R., Kranstauber, B., Carbone, C., Jansen, P.A., 2014. Quantifying levels of animal activity using camera trap data. Methods Ecol. Evol. 5 (11), 1170–1179.
Royle, J.A., 2004. N-mixture models for estimating population size from spatially replicated counts. Biometrics 60 (1), 108–115.
Schofield, G., Bishop, C.M., Katselidis, K.A., Dimopoulos, P., Pantis, J.D., Hays, G.C., 2009. Microhabitat selection by sea turtles in a dynamic thermal marine environment. J. Anim. Ecol. 78, 14–21.
Searle, C.E., Bauer, D.T., Kesch, M.K., Hunt, J.E., Mandisodza-Chikerema, R., Flyman, M. V., Loveridge, A.J., 2020. Drivers of leopard (Panthera pardus) habitat use and relative abundance in Africa's largest transfrontier conservation area. Biol. Conserv. 248, 108649.
Steenweg, R., Hebblewhite, M., Kays, R., Ahumada, J., Fisher, J.T., Burton, C., Brodie, J., 2017. Scaling-up camera traps: monitoring the planet's biodiversity with networks of remote sensors. Front. Ecol. Environ. 15 (1), 26–34.
Tanwar, K.S., Sadhu, A., Jhala, Y.V., 2021. Camera trap placement for evaluating species richness, abundance, and activity. Scientificreports 11 (1), 1–11.
Triguero-Ocaña, R., Martínez-López, B., Vicente, J., Barasona, J.A., Martínez-Guijosa, J., Acevedo, P., 2020a. Dynamic network of interactions in the wildlife-livestock interface in mediterraneanspain: an epidemiological point of view. Pathogens 9 (2).
Triguero-Ocaña, R., Vicente, J., Palencia, P., Laguna, E., Acevedo, P., 2020b. Quantifying wildlife-livestock interactions and their spatio-temporal patterns: is regular grid camera trapping a suitable approach? Ecol. Indic. 117, 106565.
Vicente, J., Soriguer, R., Gortázar, C., Carro, F., Acevedo, P., Barasona, J.A., Negro, J.J., 2014. Bases técnicas para una extracción sostenible de ungulados del Parque Nacional de Doñana. Unpublished results.
Wakefield, E.D., Phillips, R.A., Trathan, P.N., Arata, J., Gales, R., Huin, N., Matthiopoulos, J., 2011. Habitat preference, accessibility, and competition limit the global distribution of breeding Black-browed Albatrosses. Ecol. Monogr. 81 (1), 141–167.
Watanuki, Y., Suryan, R.M., Sasaki, H., Yamamoto, T., Hazen, E.L., Renner, M., Sydeman, W.J., 2016. Spatial ecology of marine top predators in the North Pacific: tools for integrating across datasets and identifying high use areas. In: North Pacific Marine Science Organization (PICES).
Wilmers, C.C., Nickel, B., Bryce, C.M., Smith, J.A., Wheat, R.E., Yovovich, V., 2015. The golden age of bio-logging: how animal-borne sensors are advancing the frontiers of ecology. Ecology 96 (7), 1741–1753.